

Vanessa Adamandia Alwan

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BIOGRAPHY

Data Scientist on the Marketing Technology (Inference) team at Airbnb. PhD in Quantitative Marketing from the University of Chicago Booth School of Business. My research focuses on applying structural and machine learning models to draw causal inference on marketing decisions, including pricing, competitive conduct, and consumer demand.

EXPERIENCE

Airbnb <i>Data Scientist – Marketing Technology (Inference)</i>	2026–Present
Duke University , Durham, NC <i>Research Assistant for V. Joseph Hotz</i>	2018–2020
University of Michigan–Ann Arbor , Ann Arbor, MI <i>Research Assistant for Anocha Aribarg</i>	2017–2018
Madison Gas and Electric , Madison, WI <i>Forecasting Intern</i>	2016–2017

EDUCATION

University of Chicago Booth School of Business Ph.D., Quantitative Marketing	Chicago, IL 2020–2026 (expected)
University of Chicago Booth School of Business M.B.A.	Chicago, IL 2020–2026 (expected)
Duke University M.S. in Economics and Computation	Durham, NC 2018–2020
University of Wisconsin-Madison B.A. in Statistics (with Honors), B.A. in Mathematics, Minor in Computer Science	Madison, WI 2014–2018

RESEARCH INTERESTS

Marketing Analytics, Causal Inference, Machine & Deep Learning, Generative and Multimodal Models, Pricing, Personalization, Privacy, Public Policy, Health and Nutrition

WORKING PAPERS

Stockpiling and the Discount Function <i>Conditionally Accepted at Marketing Science</i> with Jean-Pierre Dubé, Øystein Daljord, and Xinyao Kong Identifies the discount factor in dynamic consumer models with stockpiling behavior using nonlinear estimation and simulation.	2026
Product Attributes, Cross Elasticities and Dynamic Market Structure with Pradeep Chintagunta, Wenxi Li Model evolving market structure and consumer sensitivities using a Bayesian dynamic linear framework with attribute-space distances for cross-price elasticities, demonstrated with large-scale yogurt sales data.	2024

WORK IN PROGRESS

Greedflation and Grocery Retail Conduct: Evidence from Commodity-Like Product Markups <i>Dissertation</i> Quantifies price dispersion and cost pass-through during inflationary shocks across grocery retailers. Combines causal inference with large-scale scanner data to test for collusive pricing behavior.	2026
Cross Product Learning with Product Label Data Builds multimodal learning models integrating text embeddings from product labels with sales data to predict cross-product substitution and market dynamics.	2024

Designs privacy-preserving pricing algorithms using deep neural networks, studying efficiency tradeoffs between data protection and model accuracy.

ACADEMIC ACTIVITIES

Invited / Contributed Presentations

- “Pricing with Differential Privacy” — *Marketing Science*, Sydney, Australia (2024)
- “Two-Side Learning: Application to the Organics Markets” — *Marketing Dynamics Conference*, Santorini, Greece (2024)
- “The Identification of the Discount Factor in Dynamic Discrete-Choice Models with Stockpiling” — *Marketing Dynamics Conference*, Atlanta, GA (2022)

Invited Research Tutorials

- NBER Economics of Privacy Tutorial, Boston, MA (2022)
- ISMS Doctoral Consortium Fellow, Miami, FL (2023)

SKILLS

- **Programming:** Python, R, C++, SQL, Java, Stata, Matlab, SAS
- **Machine Learning & AI:** Deep learning (PyTorch, TensorFlow), causal ML, representation and multimodal learning, transformers, time-series forecasting, Bayesian estimation
- **Econometrics & Modelling** Structural demand estimation (BLP), GMM, panel and dynamic models, natural experiments
- **Distributed Systems:** Experience with large-scale datasets, distributed and cloud-based computation
- **Data Management:** Large-scale structured/unstructured datasets, text embedding models, panel data
- **Tools/Environments:** Linux, Git, Jupyter, RStudio
- **Communication:** Presentations at international conferences; extensive teaching and mentoring experience; skilled at translating technical results for diverse audiences.
- **Languages:** English (Native), Greek (Conversational), Spanish (Basic)

TEACHING

University of Chicago Booth School of Business

Teaching Assistant, MBA & EMBA Programs (2022–2024)

Courses: Pricing Strategies (BUSN 37802), Data-Driven Marketing (BUSN 37103/37803), Marketing Management (BUSN 37800), Data Science for Marketing Decision Making (BUSN 37105)

Duke University

Teaching Assistant, MQM Program (2019)

Course: Market Intelligence (Marketing 552Q)